

University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

LAB REPORT

Experiment Number: 02

Experiment Title: Designing a firefighting robot using flame sensors

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1. Experiment Name

Designing a Firefighting Robot Using Flame Sensors

2. Objective

The objective of this experiment is to **design and implement an automatic firefighting robot using an Arduino UNO, flame sensors, DC motors, a servo motor, and a water pump**. The robot detects the presence and direction of fire using flame sensors, moves toward the fire, and activates the water pump to extinguish it.

3. Apparatus / Hardware & Software Requirements

- List all required tools and components:

Hardware Requirements:

- Arduino UNO
- 2 × Flame Sensors | IR Sensor
- L293D Motor Driver IC
- 2 × DC Gear Motors
- 1 × Servo Motor
- 1 × Water Pump
- Relay Module / Transistor for pump control
- Buzzer
- Breadboard
- Jumper Wires
- 9V Battery / External Power Supply
- Water container
- Small pipe/nozzle for water spraying

Software Requirements

- Arduino IDE
- Tinkercad Circuits (for simulation/design)
- Arduino programming language (C/C++)

4. Circuit Diagram / Schematic

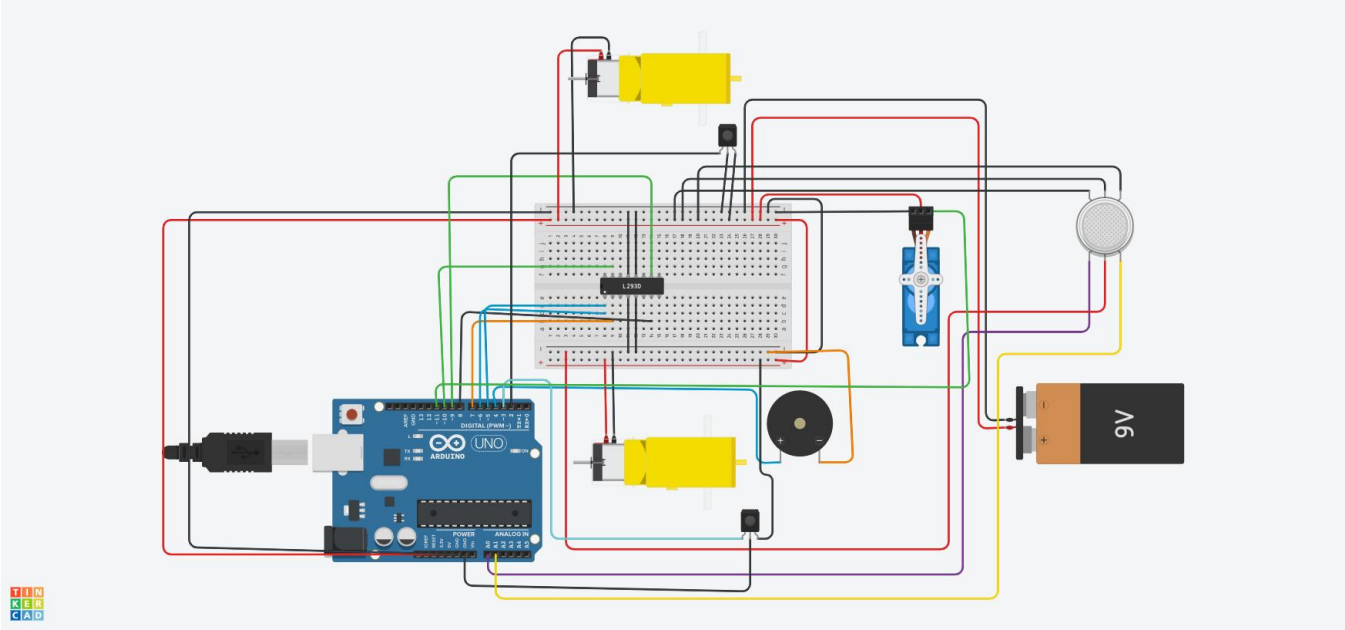


Fig: FireFighting robot

5. Code / Assembly Program

```
#include <Servo.h>

// Sensor Pins

const int leftIR = 2;
const int rightIR = 3;
const int gasSensor = A0;

// Alert

const int buzzer = 4;
const int led = 13;

// L293D Motor Driver

const int ENA = 5;
const int ENB = 6;

const int IN1 = 7;
const int IN2 = 8;
const int IN3 = 9;
const int IN4 = 10;


// Servo & Pump

const int servoPin = 11;
const int pump = 12;

Servo nozzle;

// Gas threshold
int gasThreshold = 400;


void setup()
{
  // Sensors
  pinMode(leftIR, INPUT);
  pinMode(rightIR, INPUT);

  // Alert
  pinMode(buzzer, OUTPUT);
  pinMode(led, OUTPUT);

  // Pump
```

```
  pinMode(pump, OUTPUT);

  // Motor driver
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);

  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  // Servo
  nozzle.attach(servoPin);
  nozzle.write(90);

  // Motor speed
  analogWrite(ENA, 180);
  analogWrite(ENB, 180);

  Serial.begin(9600);
}

void loop()
{
  int leftValue = digitalRead(leftIR);
  int rightValue = digitalRead(rightIR);

  int gasValue = analogRead(gasSensor);

  Serial.print("Left IR: ");
  Serial.print(leftValue);

  Serial.print("  Right IR: ");
  Serial.print(rightValue);

  Serial.print("  Gas: ");
  Serial.println(gasValue);

  // GAS / SMOKE DETECTED


  if (gasValue > gasThreshold)
  {
    Serial.println(" FIRE / SMOKE DETECTED!");

    digitalWrite(led, HIGH);
    digitalWrite(buzzer, HIGH);

    // Stop robot
```

```
  stopRobot();

  delay(500);

  // Turn ON water pump
  digitalWrite(pump, HIGH);

  // Move water nozzle
  nozzle.write(60);
  delay(500);

  nozzle.write(120);
  delay(500);

  nozzle.write(90);
  delay(1000);

  // Turn OFF pump
  digitalWrite(pump, LOW);

  digitalWrite(buzzer, LOW);
}

// NO FIRE - USE IR FOR MOVEMENT

else
{
  digitalWrite(led, LOW);
  digitalWrite(buzzer, LOW);
  digitalWrite(pump, LOW);

  // Both sensors detect line/object
  if (leftValue == HIGH && rightValue == HIGH)
  {
    forward();
  }

  // Left sensor detects
  else if (leftValue == LOW && rightValue == HIGH)
  {
    left();
  }

  // Right sensor detects
  else if (leftValue == HIGH && rightValue == LOW)
  {
    right();
  }
```

```
    // Both sensors LOW
    else
    {
      stopRobot();
    }
  }

  delay(100);
}

// MOTOR FUNCTIONS

void forward()
{
  // Left motor
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  // Right motor
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}


void backward()
{
  // Left motor
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);

  // Right motor
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}


void left()
{
  // Left motor stop
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);

  // Right motor forward
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void right()
```

```
{
  // Left motor forward
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  // Right motor stop
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}

void stopRobot()
{
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}
```

6. Output / Observations

After uploading the program, the robot behaves according to the detected flame position.

Observed Behavior:

Flame Sensor	Robot Action
No flame	Robot moves forward
Left sensor detects flame	Robot turns left
Center sensor detects flame	Robot stops and activates water pump
Right sensor detects flame	Robot turns right
Fire at center	Servo points nozzle toward fire

When the **center flame sensor detects fire**, the robot stops moving and the water pump starts spraying water. The buzzer also turns on to indicate that fire has been detected.

7. Result

The firefighting robot was successfully designed using **Arduino UNO and flame sensors**. The robot was able to detect the direction of the flame, move toward it, stop when the flame was directly in front, and activate the water pump for firefighting.

Therefore, the objective of the experiment was **successfully achieved**.

8. Conclusion

This experiment helped us understand how flame sensors, Arduino UNO, motor drivers, DC motors, servo motors, and water pumps can be integrated to build an automatic firefighting robot. We learned how sensor input can be used to control the movement of a robot and activate an actuator automatically. The project also provided practical experience in Arduino programming, circuit connections, sensor interfacing, and basic robotics.